



On the Emancipation of PLS-SEM: A Commentary on Rigdon (2012)



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Rigdon's (2012) thoughtful article argues that PLS-SEM should free itself from CB-SEM. It should renounce all mechanisms, frameworks, and jargon associated with factor models entirely. In this comment, we shed further light on two subject areas on which Rigdon (2012) touches in his discussion of CB-SEM and PLS-SEM. Rigdon (2012) highlights ways to make better use of PLS-SEM's predictive capabilities, for example, by reverting to set correlations. We discuss this issue in more detail, highlighting the need to examine the predictive capabilities of models when developing and testing theories, and broach the issue of confirmatory versus exploratory modeling. As a result of our discussion, we call for the continuous improvement of the PLS-SEM method to uncover its capabilities for theory testing while retaining its predictive character.

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Introduction

Partial least squares structural equation modeling (PLS-SEM) has become an increasingly visible methodological approach in business research. Several review studies document its increasing use across a variety of disciplines (Hair et al., 2012c; Hair et al., 2012a; Ringle et al., 2012). In addition, *Long Range Planning*, one of the leading journals in the strategic management field, has devoted three special issues to the method (Hair et al., 2012b, 2013; Robins 2012, 2014; Sarstedt et al., 2014), providing a clear indication of the importance of PLS-SEM for research and practice.

As with any development in research, the proponents and critics of PLS-SEM sometimes have heated debates on the method's advantages and disadvantages (e.g., Goodhue et al., 2012; Marcoulides et al., 2012; Henseler et al., 2014), disagreeing on whether it should be increasingly used or be applied at all. While an outright rejection of any method is certainly not good research practice and is unfounded in light of PLS-SEM's manifold advantageous features (e.g., Hair et al., 2011; Henseler et al., 2009), almost all methodological studies provide a balanced and constructive perspective on PLS-SEM's capabilities and limitations (e.g., Jöreskog and Wold 1982; Reinartz et al., 2009).

Nevertheless, much of the discussion has centered on a comparison of PLS-SEM with its longer established sibling: covariance-based structural equation modeling (CB-SEM). While these comparisons of statistical methods are important to learn more about situations that favor the use of one method over the other, we believe that this research stream has reached a point that requires a different angle and new arguments to pursue.

Rigdon's (2012) thoughtful article takes the first important step in this respect by arguing that PLS-SEM should free itself from CB-SEM. It should renounce all mechanisms, frameworks, and jargon associated with factor models entirely. We fully agree with the spirit of this appeal but also acknowledge that this step is likely to trigger resistance from authors, reviewers, and editors because SEM is grounded in decades of psychometric research.

In this comment, we shed further light on two subject areas on which Rigdon (2012) touches in his discussion of CB-SEM and PLS-SEM.¹ Rigdon (2012) highlights ways to make better use of PLS-SEM's predictive capabilities, for example, by reverting to set correlations. While prediction is the mainstay of econometrics and all related methods, in a SEM context, it is often considered the ugly stepsister of testing causal relationships. We discuss this issue in more detail, highlighting the need to examine the predictive capabilities of models when developing and testing theories. In this context, we adopt Rigdon's

¹ Rigdon (2012) distinguishes between factor based and component-based SEM. In this comment, we refer to the statistical techniques that are mainly employed for these two different approaches to SEM. Hence, the term CB-SEM is used to refer to factor-based SEM, and PLS-SEM represents component-based SEM, because it is regarded as the "most fully developed and general system" of component-based SEM techniques (McDonald, 1996; p. 240).

(2012) notion that “researchers must develop an entirely different approach to measure validation.” (Rigdon 2012, p. 354). By this means, we broach the issue of confirmatory versus exploratory modeling, the latter of which is – despite contrary notions – the dominant modeling approach in SEM. As a result of our discussion, we call for the continuous improvement of the PLS-SEM method to uncover its capabilities for theory testing while retaining its predictive character.

Prediction and explanation

Rigdon (2012) argues that PLS-SEM should fully emancipate itself from CB-SEM by stressing its prediction-orientation rather than aiming at testing model relationships in an explanatory sense (i.e., theory testing). However, is there really a dichotomy between predictive and explanatory modeling?

In strategic management and other social sciences disciplines, statistical methods are predominantly used for explanation (i.e., theory testing). The goal is to apply data in order to test hypotheses relating to relationships embedded in a nomological net. Based on statistical inference, conclusions are drawn about the tenability of the causal hypotheses, effect sizes, and the appropriateness of the entire model. In contrast, when the focus is on prediction, the goal is to predict the output values through the input values or new observations.

The concept of prediction originates from an econometric perspective and is defined as “the estimate of an outcome obtained by plugging specific values of the explanatory variables into an estimated model” (Wooldridge 2003; p. 842). In the context of SEM, Bagozzi and Yi (2012) argue that prediction relates to a situation where a theory leads to the forecast of some relevant outcome. Specifically, they note the following: “If a study tests a theory and exogenous and endogenous variables are linked significantly according to the theory, we might term the relationship *explanatory* (e.g., ξ explains η) and then decide whether or not, or to what degree, causality can be claimed. [...] Nevertheless, we prefer to use the term *prediction* when an existing theory leads to the forecast or discovery of a new phenomenon or outcome. This latter usage is consistent with some philosophy of science characterizations of what constitutes a (strong) theory. That is, a theory that explains what it is supposed to explain is given less acclaim than one that also leads to new discoveries or predictions.” (Bagozzi and Yi 2012, p. 23). This notion is also underlined by Colquitt and Zapata-Phelan’s (2007) taxonomy of theoretical contributions of empirical articles whose conceptualization captures the degree to which predictions are grounded in logical speculation or existing theory. Similarly, Roberts and Pashler (2000; p. 359) state that “a prediction is a statement of what a theory does and does not.” Jointly, these statements underline that prediction is an integral part of theory assessment, suggesting that researchers should not blindly rely on the explanation of relationships among constructs, but also keep the predictive capabilities of their model in mind (Bacharach 1989).

Unfortunately, the actual use of the SEM methodology (and others) does not adequately reflect this notion as researchers routinely neglect the importance of prediction. For example, in their analysis of more than 1,000 articles published in the *MIS Quarterly* and *Information Systems Research* journals between 1990 and 2006, Shmueli and Koppius (2010) identified only 52 empirical papers that focus on prediction. Several decades ago, the very same concern had been raised by Herman Wold, the inventor of the PLS-SEM method. He felt that SEM research generally focused too heavily on estimation and description at the expense of prediction (Dijkstra, 2010). However, the problem might even have deeper roots in that researchers are not fully aware of the distinction between prediction and explanation. As Shmueli (2010, pp. 289–290) points out: “The lack of a clear distinction within statistics has created a lack of understanding in many disciplines of the difference between building sound explanatory models versus creating powerful predictive models, as well as confusing explanatory power with predictive power. The implications of this omission and the lack of clear guidelines on how to model for explanatory versus predictive goals are considerable for both scientific research and practice and have also contributed to the gap between academia and practice.”

We believe that the frequent neglect of predictive modeling and/or the confounding of explanatory and predictive modeling in the social sciences disciplines are the sources of misunderstandings and misapplications of the SEM methods. CB-SEM has been designed for explanation, not prediction. In fact, the factor indeterminacy problem makes factor-based methods such as CB-SEM hardly suitable for predictive modeling and prediction-oriented research (Rigdon, 2012). In contrast, PLS-SEM’s prediction orientation was recognized very early on as one of the method’s major strengths (Jöreskog and Wold, 1982). The extraction of latent variable scores, in conjunction with the method explaining a large percentage of the variance in the indicator variables, is useful for accurately predicting latent variable scores (Anderson and Gerbing, 1988). PLS-SEM’s superiority in terms of prediction has also been validated in a simulation study by Reinartz et al. (2009, p. 340). They conclude that “PLS is preferable to ML-based CBSEM when the research focus lies in identifying relationships (i.e., prediction and theory development) instead of confirming them.”

Correspondingly, prediction orientation has become one of the most prominently cited reasons for preferring PLS-SEM over CB-SEM. This is evidenced in practically all reviews of PLS-SEM use across different disciplines (e.g., Hair et al., 2012c; Hair et al., 2012a; Ringle et al., 2012). However, these reviews also reveal that rather than fully subscribing to predictive modeling, PLS-SEM researchers frequently frame their results as reporting in a confirmatory sense. Instead, researchers should broaden their focus and also consider prediction as an important analysis goal. It is, after all, in the very nature of business research (as an applied discipline) to examine levers with which to predict improvements in company performance well, and thereby to provide recommendations for decision-making (Diamantopoulos et al., 2012). Correspondingly, instead of mimicking the CB-SEM methodology, PLS-SEM should truly serve predictive modeling purposes.

Assessment of results

An important element of the predictive modeling process relates to the evaluation of results. In his article, Rigdon (2012, p. 353) concludes that “the PLS path modeling community should work to complete and validate a purely composite-based approach to evaluating modeling results.” This call implies that the existing PLS-SEM evaluation criteria are incomplete, not purely composite-based, or both. But what should PLS-SEM-based evaluation criteria achieve? From our perspective, criteria should provide answers to the following two questions: (1) Is the model able to explain and/or predict the dependent variable(s)?, and (2) Does the model adequately explain the observed correlations between variables? The first question refers to the use of adequate evaluation criteria for PLS-SEM, while the second question addresses the issue of model specification.

PLS-SEM evaluation criteria

The PLS-SEM toolbox includes a broad range of evaluation criteria to assess the adequacy of the measurement and structural models as described in the extant literature (Chin 1998, 2010; Hair et al., 2014; Hair et al., 2013; Hair et al., 2012a). However, there is a scarcity of clear, prediction-focused criteria.

The coefficient of determination (R^2) is typically used as a criterion of predictive power (Hair et al., 2012a; Henseler et al., 2011; Sarstedt et al., 2013). However, the R^2 only has informative value with regard to in-sample prediction. In contrast, as a measure of predictive relevance, Stone-Geisser's Q^2 (Geisser 1974; Stone 1974) provides a gauge for out-of-sample prediction. In the structural model, a Q^2 value larger than zero for a particular reflective endogenous latent variable indicates the path model's predictive relevance for this particular construct. In contrast, a Q^2 value smaller than zero indicates that the model does not perform better than the simple average of the endogenous variable would do. It should, however, be noted that while comparing the Q^2 value to zero is indicative of whether an endogenous latent variable can be predicted, it does not say anything about the quality of the prediction. In analogy to the f^2 effect size, researchers can compute the q^2 effect size, which allows for evaluating the relative impact of one construct in terms of its predictive relevance (Chin 1998; Hair et al., 2014).

Even though Herman Wold recognized the usefulness of Stone-Geisser's Q^2 early on by stating that it fits PLS-SEM “like hand in glove” (Wold 1982, p. 30), this criterion is seldom reported in PLS-SEM studies (Hair et al., 2012c; Hair et al., 2012a; Ringle et al., 2012). Furthermore, if reported, it is usually to be found in a results table without any critical analysis and further interpretation. Hence, the PLS-SEM community needs to better understand the use of suitable predictive evaluation criteria.

Relying on such well-known measures is not sufficient. PLS-SEM's toolbox has been and must be further extended to improve the predictive capabilities of the model estimation. For example, recent improvements address the critical issue of unobserved heterogeneity, which threatens the validity of all SEM results (Becker et al., 2013; Sarstedt 2008a, b) and should become a standard means of evaluating PLS-SEM results (Hair et al., 2012a). Newly developed PLS-SEM segmentation methods – such as finite mixture partial least squares (FIMIX-PLS; Hahn et al., 2002; Sarstedt et al., 2011; Sarstedt and Ringle 2010), the prediction-oriented segmentation of PLS-SEM results (PLS-POS; Becker et al., 2013), or genetic algorithm segmentation (PLS-GAS; Ringle et al., 2014; Ringle et al., 2013) – can assist researchers conducting this kind of analysis. Other directions for extending the toolbox include PLS-SEM's ability to extract multiple dimensions from a given path model and set of indicators (e.g., Kuppelwieser and Sarstedt 2014). This is also noted by Rigdon (2012, p. 354): “PLS path modeling typically forms just one composite for each set of indicators, constraining the relationship between each set of predictors to be unidimensional. Cohen's (1982) set correlation accounts for the explanatory power of relationships between sets of predictors across multiple dimensions. Comparing a particular model with a set correlation analysis would show just how much predictive ability a researcher was ‘leaving on the table’ for the sake of specifying unidimensional relationships.” A corresponding extension has already been introduced by Apel and Wold (1982). Their deflation technique estimates latent variables in two and more dimensions by using the residuals of the previous dimension as the PLS-SEM algorithm's new input to obtain the estimations of the next dimension.

To summarize, PLS-SEM must overcome several problems before its value can be fully understood. First, further criteria and evaluation techniques for PLS-SEM (e.g., the use of root mean squared covariances, Tucker-Lewis and Bentler-Bonett indices, deflation) need to be considered. Second, PLS-SEM research should make use of existing criteria and develop further PLS-SEM-specific evaluation criteria and complementary analysis techniques that stress the method's predictive character. We expect lively academic discussions and increased numbers of research publications on PLS-SEM improvements in line with establishing a predictive modeling process tailored to PLS-SEM.

Model misspecification

We fully agree with Rigdon's (2012) call that PLS-SEM should play out its strengths as a distinctive prediction-oriented approach to SEM with its own set of suitable evaluation criteria. However, we also believe that this notion should not mean that methodological research can lean back and stop seeking for solutions to the limitations of the method, such as its inability to detect model misspecification.

Philosophy of science tells us that valid conclusions can only be drawn from a system if its separate assumptions are true. For SEM – whether factor-based or component-based – it is therefore crucial to identify and eliminate model misspecification

in structural equation models (Hu and Bentler, 1998). Hence, in addition to recognizing the situations where PLS-SEM has unique advantages over CB-SEM by further developing a set of suitable prediction-oriented evaluation criteria, we should also focus on answering the question whether the path model adequately explains the observed correlations between variables in order to avoid model misspecification.

The current guidelines for model evaluation have limited value in detecting model misspecification. In particular, none of the evaluation criteria recommended in extant PLS-SEM tutorials (e.g., Hair et al., 2014; Henseler et al., 2012; Henseler et al., 2009) are able to identify problems of underparameterization. This is due to PLS-SEM, which unlike CB-SEM, lacks a global scalar function that could be used as an indicator of whether or not a model fits the data. In fact, the term fit has different meanings in the context of CB-SEM and PLS-SEM. Whereas in CB-SEM, fit refers to the distance between an observed covariance matrix and an implied covariance matrix, in PLS-SEM fit relates to the degree to which a correlation or covariance-based criterion is being maximized (Hanafi, 2007; Tenenhaus and Tenenhaus, 2011). Correspondingly, “goodness-of-fit” measures in PLS-SEM such as those introduced by Tenenhaus et al. (2005) (i.e., GoF and GoF_{rel}), can – by definition – not offer what their names promise. Henseler and Sarstedt (2013, p. 577) conclude: “Neither of these indices is able to separate valid models from invalid models. In fact, researchers would be misled if they chose the model yielding the highest GoF.”²

A starting point for developing model fit measures that are capable of detecting misspecification could lie in Lohmöller's (1989) work. More than 20 years ago, he suggested that three evaluation criteria should be used to examine the adequacy of PLS path models. Lohmöller stressed that the outer residual correlation matrix as well as the inner residual correlation matrix may indicate that the model cannot fully explain the relationships between indicators, or may indicate that the relations included in the structural model do not fully explain the interplay of the constructs. The standardized root mean square residual (SRMR), which represents the Euclidean distance between the empirical correlation matrix and the model-implied correlation matrix, is a third instrument with which to detect model underestimation (Hu and Bentler, 1998; Jöreskog, 1993). The SRMR was initially proposed for use in combination with CB-SEM (Jöreskog and Sörbom, 1981) but has also been transferred to PLS-SEM (Lohmöller 1989).³ Unfortunately, the extant literature has not adapted any of these measures for model evaluation in PLS-SEM or developed guidelines for their use (e.g., developing thresholds).

While evaluation criteria can help detect model misspecification, they are not sufficient to avoid misspecification. Currently, most variance-based SEM techniques have methodological limitations that restrict analysts' flexibility to specify models. For instance, most variance-based SEM techniques assume recursive models, which means that researchers cannot model feedback loops or endogeneity. Consequently, researchers using variance-based SEM can hardly avoid misspecification if the true model is not recursive. Researchers should develop ways to overcome the necessity for recursive path models. Only recently, Henseler et al. (2014) empirically demonstrated its usefulness for model validation purposes.

It's all about model fit – really?

Following the route to model fit also has risks, as can be seen in most CB-SEM applications. When using CB-SEM, it is very common for initially hypothesized models to exhibit inadequate fit. In response to this issue, researchers should reject the model and re-consider the study (which usually requires gathering new data). Alternatively, they frequently re-specify the original theoretically developed model in an effort to improve fit indices beyond the suggested threshold levels. By doing so, researchers arrive at a well-fitting model, which they conclude theory supports. Unfortunately, the latter is a best-case scenario that almost never applies in reality. Rather, researchers engage in exploratory specification searches for model set-ups that yield satisfactory levels of model fit.

In this context, Diamantopoulos (1994, p. 123) stresses that “the nature of the analysis is no longer confirmatory (i.e., testing a pre-determined system of hypotheses as reflected in the original model specification) but becomes exploratory in nature.” As models resulting from such specification searches often capitalize on the idiosyncrasies of the sample data (e.g., Chou and Bentler, 1990; Green et al., 1998; MacCallum and Browne, 1993), “the final models that are the product of such modifications often do not correspond particularly well to the correct population models.” (Tomarken and Waller, 2003, p. 595). Correspondingly, simulation studies show that a CB-SEM specification search is of rather poor quality (Homburg and Dobratz, 1992). Furthermore, owing to the identification problems and decreased levels of statistical power when the sample size is limited (Reinartz et al., 2009; Vilares and Coelho, 2013) – a circumstance researchers face more often than not (Diamantopoulos et al., 2012) – such specification searches also tend to favor less complex models. When theories are more elaborate and path models increase in complexity, this tendency may prove problematic regarding advancing our understanding of certain phenomena (especially when using covariance-based approaches to SEM, which are clearly limited in terms of estimating complex path models). Overall, it seems reasonable to conclude that while CB-SEM is traditionally viewed as a confirmatory tool, the contrary is actually true in practice.⁴

² The same holds for the FIT, and the adjusted FIT indices (Hwang et al., 2010) of generalized structured component analysis (Henseler, 2012).

³ Note that the SRMR indices are not exactly the same for the two techniques. Different from SRMR in CB-SEM, in PLS-SEM, the discrepancies between the empirical correlations and the model-implied correlations are zero for pairs of indicators belonging to the same construct.

⁴ Researchers will have a hard time admitting the exploratory character of their CB-SEM use considering (1) the problems that emerge from ex post factor analyses (Cliff, 1983) and (2) the academic bias in favor of findings presented in confirmatory terms (Greenwald et al., 1986).

In contrast, PLS-SEM has been designed for research situations that are “simultaneously data-rich and theory-primitive” (Wold 1985, p. 589). Wold envisioned a discovery-oriented process, “a dialogue between the investigator and the computer” (Wold 1985, p. 590). Rather than a priori committing to a specific model and framing the statistical analysis as a hypothesis test, Wold expected that researchers would estimate numerous models in the course of learning something about the data and about the phenomena. It is ironic that Wold’s vision currently applies to many allegedly confirmatory CB-SEM analyses rather than many exploratory PLS-SEM analyses.

Therefore, instead of fully following the model fit maximization paradigm of explanatory modeling, the goal of predictive modeling should be to establish theoretically grounded models that have high predictive power. In this sense, methodological research should aim at uniting the strengths of both methods so that we truly arrive at “Not CB-SEM versus PLS-SEM” but “CB-SEM and PLS-SEM” (Hair et al., 2012a; p. 415).

Conclusion

Bentler and Huang’s (2014) as well as Dijkstra’s (2014) comments in this issue of *Long Range Planning* introduce corrections of PLS-SEM estimates that provide a method to mimic CB-SEM results perfectly. If these approaches deliver what they promise, PLS-SEM is capable of delivering a comparable result to CB-SEM – if not of the same precision – but keeps most of its advantageous features (e.g., use of complex models, modeling flexibility, relatively low demands regarding data distribution and sample size, convergence behavior of the algorithm, and stability of results). These advances not only prove most arguments of PLS-SEM critics wrong, but also allow for adopting techniques and evaluation criteria that have been developed and established for CB-SEM in the past decades.

Exploiting the explanatory abilities of PLS-SEM for theory testing and emancipating the method by further developing its predictive capabilities will allow researchers to address both analytical concerns (i.e., explanation and prediction) by using a single method. In line with these considerations, the corrected and further extended PLS-SEM methods may be used in a combined explanatory and predictive modeling process, thereby constituting the third generation of multivariate analysis for strategic management and other social sciences disciplines (on the second generation of multivariate analysis see Fornell, 1982). SEM research and applications in general – as one of the most important multivariate analysis techniques of the social sciences disciplines – will benefit from these fundamental developments.

Despite these advancements, PLS-SEM should retain its predictive character rather than fully subscribing to explanatory modeling. For this purpose, a predictive modeling procedure specific to PLS-SEM should be established that highlights the specifics of predictive modeling in terms of model building and assessment (Shmueli, 2010). Regarding the latter, researchers should make better use of established criteria and introduce new ones to assess the predictive capabilities of their models. Moreover, PLS-SEM should use its existing procedures properly to avoid model misspecification.

Lastly, from a more generic academic research perspective, the recent exchange on Rigdon’s (2012) article shows the benefits of a constructive discussion when dealing with controversial topics such as the “To PLS or Not to PLS” debate. Any extreme position that (oftentimes systematically) neglects the beneficial features of the other technique, and may result in prejudiced boycott calls, is not good research practice and does not help to truly advance our understanding of methods and any other research subject (Hair et al., 2012b).

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